Name: ______________________________ **Date:** _________________ **Class:** _______________

MyDesign Portfolio

Element I Initial Template

How to use this template

You must test *all* high priority requirements to score a 5. After testing, you will connect the data you collected to the "success" or "failures" of your design. You will consider next steps for your design based on the data you just analyzed. Each test that was listed in your table for Element H (testing plan) was on its own row. Generate a report for each test, the results, and its analysis that includes the following:

a) number and name of the test (as referenced in Element H)
b) indication of whether the test is high, medium, or low priority based on your requirements (as reference in Element C)
c) for physical tests, include an explanation of your process and the full set of raw data such that the following best practices for scientific testing and mathematical modeling will be evident:
    - creating experimental controls
    - defining independent and dependent variables
    - only changing one variable at a time and changing it in increments to find a trend
    - collecting at least FIVE trials for each variable increment
    - plotting data appropriately (e.g. scatter plot with trendline for continuous variables, bar charts/histogram for non-continuous variables, acknowledging and recording error as appropriate, and labeling random and systematic errors).
d) summary of your data with appropriate charts, graphics, spreadsheets, etc., drawing on best practices in science and math for data display and analysis
e) statement of what the data means and a summary of the implications of the data you collected
    - A possible sentence could be, “Based on our testing data, we found that <trend in data> means <something about the design's success> because <logic of conclusion>.”
f) discussions about *each* implication from the summary of your data

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g) next steps for your design and process from the new information from the test that was conducted

- Goal 3 - Important - Durability. No numerical data. Tests included:
    - Running all motors for the span of around 10 seconds.
    - Dropping and pulling on the ROV to make sure the frame stays intact.

No further changes needed. Similar tests should be run with any major changes for the same reason we chose to test the ROV in the first place.

- Goal 2 - Very Important - Maneuverability (at JCC). No numerical data.
    - Forwards and backwards movement - very fast (2 motors).
    - Side to side - non-existent, but because of our speed in turning it's not crucial, but it's a possible place to improve if we had more motors.
    - Up and down - a little slow but it's not being used nearly as much as forwards and backwards.
    - Rotational - yaw worked exceptionally well. With the 2 motors powering it it was very easy to turn, which is good because we are really reliant on it. Roll and pitch stayed very stable because of the way the floaties were distributed.

No further changes needed, very little to improve on with current resources.

- Goal 1 - Very Important - Microplastic collection (in the Bay). No numerical data (yet).
    - Initial testing was hard to tell. It seemed very likely that much of the same was lost because of the turning movement, which had water flow through portions of the mesh backwards, washing out samples.
    - Our second test was much more promising. Taking into account our first test results we were much more careful, especially when removing the ROV from the water. Microplastics were hard to see but were picked up off the filter. We are unsure of how much as of now.

We need to be cautious while driving the ROV as to not wash out our samples. Otherwise no physical changes need to be immediately made, especially with the time constraints, but this would be an issue to look into for further revisions.

- Goal 4 - Not too important - Controls (in the Bay). No numerical data.
    - Both me and elvie tested the controls, and were able to maneuver the ROV.

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- Forwards backwards was swapped from what is intuitive (press back to go forwards). Not much to get used to since it's similar to tilting a joystick back to go up. We both found the controls easy to use after a few seconds to get used to them.

Possibly a joystick could work better? It would help with the confusion of how to turn. We also should flip the connections so that forwards and backwards are flipped if we have a chance to rewire everything.